

Student Research Internship · Stream

Machine Learning Engineer

FunkyForelle | Predicting River Oxygen to Protect Swiss Fish

About FunkyForelle

FunkyForelle is an open-source project building a universal AI layer that predicts dissolved oxygen (DO) in Swiss rivers — the variable that directly determines whether fish survive summer heatwaves. During heatwaves, when DO drops below 6 mg/L fish get stressed; below 4 mg/L, mass die-offs occur within hours. Organisations like Fischereiverein See + Gaster currently conduct fish rescue operations with only 24 hours' notice. FunkyForelle gives them 14 days.

We are extending AareML — a seq2seq LSTM developed as part of a CAS in Advanced Machine Learning at the University of Bern — into a Switzerland-wide operational DO forecasting system. The project is an open-source research project.

Your Role — Stream

You will be the sole ML engineer on Stream, building the universal DO prediction model from scratch through four monthly milestones:

- **Month 1** — Baseline model: Extend AareML to predict DO at all available FOEN river stations using temperature, flow, and water level data. Establish a reproducible training pipeline on UBELIX (Uni Bern HPC cluster).
- **Month 2** — Expanded model: Integrate Eawag Urban Water Observatory and cantonal monitoring data. Broaden coverage across Swiss rivers.
- **Month 3** — Live integration: Ingest the real-time sensor stream from Stream's field deployment. Validate DO predictions against ground-truth measurements at the pilot site.
- **Month 4** — Reserve: Stabilisation, open-source release, reproducibility guide. Prototype handed to Patrick Hager's team ahead of May 2027 summer season.

What You Will Work With

- Model: seq2seq LSTM (PyTorch), extending the open-source AareML codebase
- Data: FOEN hydrodaten API, CAMELS-CH-Chem (40 years, 115 catchments, DO included), Eawag open datasets, cantonal portals
- Compute: UBELIX HPC cluster (Uni Bern) — GPU nodes, no commercial cloud needed
- Tools: Python, PyTorch, Weights & Biases, GitHub Actions, lightweight cloud dashboard
- Expert advisor: Thiago Nascimento (Eawag) — author of CAMELS-CH-Chem, advises on data and model architecture
- Weekly check-ins with project leads Olga and Tatyana

What We Are Looking For

Required:

- Enrolled in BSc or MSc in Computer Science, Data Science, Environmental Science or related field
- Solid Python and PyTorch or TensorFlow experience
- Familiarity with time-series forecasting and ML training pipelines

- Comfortable working independently with messy real-world data

Nice to have:

- Experience with hydrology, water quality, or environmental datasets
- Knowledge of NeuralHydrology or seq2seq architectures
- Interest in open-source and responsible AI

What We Offer

- CHF 2,000/month — 4 months (40–60% FTE)
- Academic credit: we will support thesis or project course connection with your university
- UBELIX HPC access — serious compute for serious ML
- Expert mentorship from Thiago Nascimento (Eawag) and project leads
- Your model runs in production — field-tested by fishery managers from May 2027
- Co-authorship on any publications arising from the work
- Open-source contribution: code and weights published under MIT licence

Logistics

- Duration: 4 months — November 2026 to February 2027
- Location: Remote-first, with occasional meetings in Bern/Zurich
- Start date: Flexible, ideally 1 November 2026
- Open to BSc and MSc students from any Swiss or international university

How to Apply

Email polar.bear.before.lunch@gmail.com (one page max):

1. Your current program and year of study
2. Two or three sentences on why this project interests you
3. Link to your GitHub or any relevant code/project

We respond to all applications within one week.